Length-Outlier Exclusion vs. Per-Thread Subsampling

1 Problem

The synthetic comments are subsampled per thread so each (model, method) group has exactly as many comments per thread as the real group. That needs every (thread, model, method) cell to hold *at least* as many synthetic comments as the thread's real count. Removing length outliers *first* breaks this: a cell can end up with fewer synthetic comments than real ones ("shortfall"), and since many cells hold a single comment, one flagged comment can empty a cell entirely. Without any exclusion, 0 cells fall short. Table 1 lists the six rules compared.

Table 1: Length-outlier rules compared (detected within each model x method group; log = on log word count).

Rule	Comments excluded	% excluded	Cells short	% cells short
IQR1.5 raw	1262	0.98	622	1.05
IQR1.5 log	1393	1.08	453	0.76
IQR3 log	37	0.03	16	0.03
MAD3.5 log	332	0.26	105	0.18
P1-99 log	2283	1.77	975	1.64
P2.5-97.5 log	5732	4.46	2467	4.15

2 Cells that fall short, by model x method

Pooled over all groups, the rules push 0.03% (IQR3 log) to 4.15% (P2.5-97.5 log) of cells into shortfall (none without exclusion). The rule I would have provisionally chosen (MAD3.5 log) reaches up to 0.50% (Gemini / SAGII) in its worst group. Table 2 gives all groups.

3 Non-zero ratings lost, by model x method and metric

Empathy labels are discrete (0 = none; > 0 = some). The tables give the percent of each group's *non-zero-rated* synthetic comments that a rule would exclude - i.e. how much of the informative signal is lost. Pooled over all groups the loss ranges 0.08% (IQR3 log) to 7.60% (P2.5-97.5 log) for Emotional Reactions,

Table 2: Percent of (thread, model, method) cells with fewer synthetic comments than real ones after exclusion.

Group	None	IQR1.5 raw	IQR1.5 log	IQR3 log	MAD3.5 log	P1-99 log	P2.5-97.5 log
Gemini / DG	0	0.59	0.02	0.00	0.00	1.71	4.12
Gemini / SAG	0	0.77	1.29	0.03	0.17	1.03	3.08
Gemini / SAGII	0	0.71	1.77	0.11	0.50	1.00	3.46
Openai / DG	0	0.09	0.00	0.00	0.00	2.12	5.42
Openai / SAG	0	0.45	0.74	0.00	0.08	1.36	3.41
Openai / SAGII	0	1.05	0.67	0.02	0.06	1.76	4.36
Grok / DG	0	2.40	0.18	0.00	0.03	1.96	4.55
Grok / SAG	0	1.41	1.06	0.06	0.32	1.58	3.80
Grok / SAGII	0	1.96	1.14	0.03	0.44	2.26	5.18
All synthetic	**0**	**1.05**	**0.76**	**0.03**	**0.18**	**1.64**	**4.15**

0.00% (IQR3 log) to 7.39% (P2.5-97.5 log) for Interpretations, and 0.20% (IQR3 log) to 6.56% (P2.5-97.5 log) for Explorations. The loss is not evenly spread: under MAD3.5 log the worst group loses 1.23% (Grok / SAGII), 0.32% (Grok / SAGII), and 1.96% (Gemini / SAGII) of its non-zero ratings (Emotional Reactions, Interpretations, Explorations). Real comments are never excluded.

Table 3: Emotional Reactions: percent of non-zero-rated synthetic comments excluded.

Group	None	IQR1.5 raw	IQR1.5 log	IQR3 log	MAD3.5 log	P1-99 log	P2.5-97.5 log
Gemini / DG	0	1.26	1.49	0.00	0.57	3.44	7.23
Gemini / SAG	0	3.10	3.18	0.00	0.24	4.40	9.14
Gemini / SAGII	0	2.76	2.31	0.36	0.89	3.20	7.29
Openai / DG	0	0.36	0.00	0.00	0.00	2.17	3.99
Openai / SAG	0	1.29	1.58	0.00	0.00	3.01	8.31
Openai / SAGII	0	1.68	1.40	0.00	0.00	3.08	5.74
Grok / DG	0	3.45	0.00	0.00	0.00	2.30	4.60
Grok / SAG	0	3.79	3.79	0.00	0.95	5.69	10.90
Grok / SAGII	0	1.23	3.70	0.00	1.23	4.94	9.88
All synthetic	**0**	**2.16**	**2.08**	**0.08**	**0.40**	**3.54**	**7.60**

Table 4: Interpretations: percent of non-zero-rated synthetic comments excluded.

Group	None	IQR1.5 raw	IQR1.5 log	IQR3 log	MAD3.5 log	P1-99 log	P2.5-97.5 log
Gemini / DG	0	2.16	0.20	0	0.00	5.70	10.61
Gemini / SAG	0	5.17	1.53	0	0.19	5.36	9.96
Gemini / SAGII	0	6.08	2.07	0	0.28	6.08	12.71
Openai / DG	0	0.00	0.00	0	0.00	2.92	5.43
Openai / SAG	0	1.20	0.58	0	0.05	2.27	5.35
Openai / SAGII	0	1.67	0.63	0	0.00	1.99	4.71
Grok / DG	0	4.79	1.37	0	0.23	5.32	10.04
Grok / SAG	0	6.16	1.04	0	0.12	3.74	8.35
Grok / SAGII	0	3.93	1.08	0	0.32	3.23	7.27
All synthetic	**0**	**3.21**	**0.88**	**0**	**0.13**	**3.51**	**7.39**

Table 5: Explorations: percent of non-zero-rated synthetic comments excluded.

Group	None	IQR1.5 raw	IQR1.5 log	IQR3 log	MAD3.5 log	P1-99 log	P2.5-97.5 log
Gemini / DG	0	0.00	2.78	0.10	1.75	2.37	6.39
Gemini / SAG	0	0.77	3.53	0.11	0.66	2.09	5.29
Gemini / SAGII	0	1.57	5.74	0.52	1.96	2.61	8.49
Openai / DG	0	0.00	0.00	0.00	0.00	1.00	2.99
Openai / SAG	0	1.22	1.22	0.00	0.00	3.66	10.98
Openai / SAGII	0	1.40	0.70	0.00	0.00	2.80	7.69
Grok / DG	0	0.00	1.65	0.00	0.00	1.65	3.70
Grok / SAG	0	0.81	4.07	0.27	1.08	4.61	9.49
Grok / SAGII	0	0.00	2.54	0.42	1.69	2.54	5.08
All synthetic	**0**	**0.64**	**3.32**	**0.20**	**1.17**	**2.50**	**6.56**

4 Why the Bayesian models do not need this step

1. **The outcome cannot have outliers in the usual sense.** Outlier removal protects against heavy-tailed *outcome* noise. Here the outcome is a bounded category (0/1/2, or 0/1 for Interpretations and Explorations) under a Bernoulli or cumulative-logit likelihood, so no single comment can contribute an unbounded residual. "Outlier" can only mean an unusual *covariate* value, comment length.
2. **Length is modeled, not ignored.** `log(length)` is a covariate in every model, so unusual lengths are absorbed by the length slope rather than distorting the group contrasts, and the log transform compresses the long tail so no comment gets extreme leverage. The rules above touch only 0.03% to 4.46% of synthetic comments, a small share of the data, which limits how much influence they can have either way.

3. **Priors and pooling limit any small set of points.** Weakly informative priors on all coefficients and the `post_id` random intercept's shrinkage toward the population mean cap how far a handful of comments can pull an estimate.
4. **Exclusion is not neutral.** Dropping comments on a variable associated with the outcome removes non-zero ratings unevenly across groups (tables above), which can bias exactly the group contrasts we care about. That cost is real, while the protection it buys is small (points 1-3).

Robustness here comes from the model's structure (bounded categorical likelihood, length adjustment, priors), not from Bayesian inference in general - a Gaussian model of a continuous outcome would need a robust likelihood instead. If desired, the claim can be checked directly with a sensitivity fit with and without the excluded comments; that is cheap for Interpretations and Explorations, slow for Emotional Reactions.

5 Per-thread subsampling is not neutral either

Keeping a random `min(n, real count)` synthetic comments per cell means the whole analysis rests on one arbitrary draw. To see how much that matters, the Interpretations and Explorations models (binomial `glmer` versions of the Bayesian models, which are fast enough to repeat) were refit on 50 independent balanced subsamples, with no outlier exclusion. Emotional Reactions was not assessed: its ordinal fits are too slow to repeat.

Table 6 compares, for each contrast, the spread of its estimate across subsamples (SD) with the standard error of a single fit (SE). If the SD were small relative to the SE, the draw would not matter. Instead, the median SD/SE is 0.27 to 0.47 across contrast sets, and a single draw adds up to 33% to the variance of the most affected contrast (relative to that fit's own SE).

Table 6: Spread of estimates across balanced subsamples relative to a single fit's standard error.

Metric	Contrast set	N	Median SD/SE	Max SD/SE	Max var. infl.	Unstable sig.
Explorations	vs. real	9	0.35	0.52	1.27	0
Explorations	Models (in method)	9	0.47	0.57	1.33	2
Explorations	Methods (in model)	9	0.42	0.49	1.24	2
Interpretations	vs. real	9	0.27	0.46	1.21	0
Interpretations	Models (in method)	9	0.44	0.53	1.28	2
Interpretations	Methods (in model)	9	0.41	0.46	1.21	5

The headline group-vs-real contrasts never change nominal significance across draws (0 of 18 unstable). The simple-effects contrasts are less safe: 11 of 36 reach nominal significance ($|z| > 1.96$) in some draws but not others (Table 7), so whether they look significant partly depends on which comments happened to be kept.

Table 7: Simple-effects contrasts whose nominal significance depends on the subsample drawn.

Metric	Contrast set	Within	Contrast	Mean est. (log OR)	Mean SE	% draws sig.
Explorations	Models (in method)	SAGII	gemini - openai	0.54	0.17	98
Explorations	Models (in method)	SAG	openai - grok	0.11	0.20	2
Explorations	Methods (in model)	Openai	sag - sagii	-0.47	0.19	88
Explorations	Methods (in model)	Openai	dg - sagii	-0.42	0.17	86
Interpretations	Models (in method)	SAGII	gemini - openai	0.25	0.09	94
Interpretations	Models (in method)	DG	gemini - openai	0.17	0.08	48
Interpretations	Methods (in model)	Gemini	dg - sag	0.18	0.09	62
Interpretations	Methods (in model)	Openai	sag - sagii	0.10	0.05	48
Interpretations	Methods (in model)	Gemini	sag - sagii	-0.17	0.09	46
Interpretations	Methods (in model)	Openai	dg - sagii	0.09	0.05	22
Interpretations	Methods (in model)	Grok	dg - sag	0.09	0.06	20

5.1 Weighting instead of subsampling

A deterministic alternative keeps *all* synthetic comments and gives each weight `min(1, real count / cell size)` (real comments weight 1), so the weights sum to exactly the subsampled n (77,603). The expected

subsampled log-likelihood over the random draw equals this weighted log-likelihood, so a single weighted fit does not depend on any one draw and uses all 128,653 synthetic comments rather than the 77,603 a subsample keeps. The same models were fit with these weights and compared with the subsample results (Table 8).

Table 8: Weighted full-data fit vs. the average of the balanced subsamples: difference in estimates (in single-fit SEs), ratio of standard errors, and agreement in nominal significance.

Metric	Contrast set	Median |diff| (SE)	Max |diff| (SE)	Median SE ratio	Sig. agrees
Explorations	vs. real	0.42	2.95	1.11	9 / 9
Explorations	Models (in method)	1.97	3.43	1.11	8 / 9
Explorations	Methods (in model)	0.44	1.41	1.10	7 / 9
Interpretations	vs. real	1.48	2.05	1.04	9 / 9
Interpretations	Models (in method)	0.64	0.98	1.14	7 / 9
Interpretations	Methods (in model)	0.55	1.51	1.13	9 / 9

The weighted estimates sit a median of 0.65 SE from the subsample average and agree on nominal significance for 49 of 54 contrasts, but they are **not** a drop-in replacement: 7 contrasts differ by more than 2 SE (5 of them Explorations), the largest gap being 3.4 SE. The direction is not uniform: among those 7, the subsample estimate is larger in magnitude for 5. One possible explanation is rare-event bias inflating estimates in smaller samples (few non-zero ratings), but this was not tested. The weighted models' standard errors are also naive (`glmer` treats weights as precision weights, no sandwich correction) and come out about 12% larger than a single subsample's.

5.2 Implication

- Subsampling has a measurable effect: one random draw adds substantial variance and can flip the significance of borderline simple-effects contrasts, so a single draw should not be the primary analysis.
- Weighting all comments is a reasonable primary alternative - it is deterministic and uses all the data - but it is a different estimator, with differences of up to a few SEs from the subsample average for some contrasts. It should be adopted knowingly, not assumed equivalent.
- A weighted Bayesian fit would need its own check (`brms` weights give a pseudo-likelihood, so its posterior widths are approximate) and, for Emotional Reactions, could not be verified against repeated subsamples. Reporting both a weighted primary fit and a subsample sensitivity check on the cheaper mechanisms would make the choice transparent.